Mesh Water Uptake and Percent Openness Tests

As shown in Figure 3.3 with empty mesh packet load cell results, tests have been conducted to determine optimal mesh materials. **Figure 3.6** shows results of mesh sample water uptake tests. The first mesh, MM40 is a polyester material with a pore size of 40 microns, while sample A25 is a nylon material and has 25-micron pores. A25 has been the standardized mesh for all pixie stick packets thus far. Another sample of A25 mesh was tested after treating with Starbrite hydrophobic coating. MM40 has a much lower water uptake than A25; however, the Starbrite coating reduced water uptake in A25 significantly. This matches data trends in Figures 3.3 and 3.5.

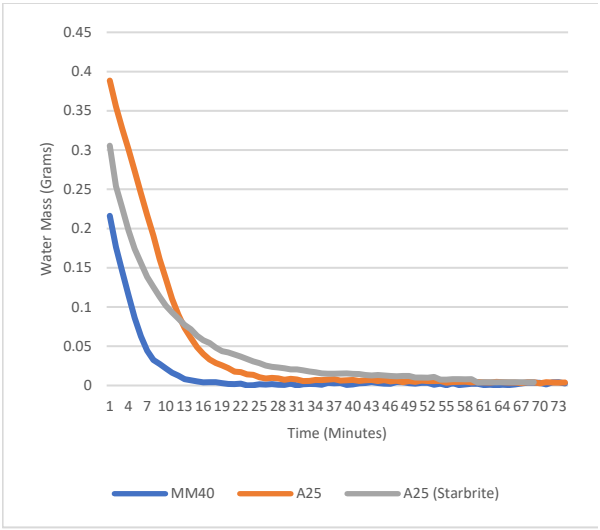


Figure 3.6. Water uptake in soak tests of blank mesh sheets. Samples include MM40 (40-micron pore size polyester with 25% openness), and A25 (25-micron pore size nylon with 8% openness) with and without Starbrite coating.

A critical parameter for ensuring rapid CO_2 delivery is the mesh pore size and the ratio of open pores (empty space) versus closed cloth of the mesh, also known as percent openness. To determine percent openness of samples MM40, A25 and A25 with

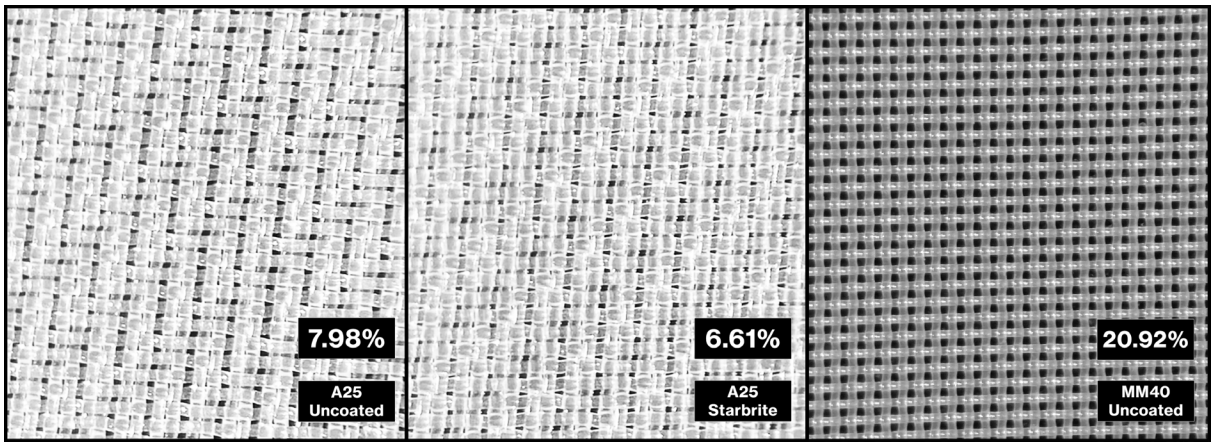


Figure 3.7. Mesh samples of A25 before and after Starbrite Coating and uncoated MM40. Percent openness is calculated as the ratio of open pore to closed fabric of the mesh.

Starbrite coating, photos were taken at high magnification and then processed to create high contrast black and white images with high resolution. Using photoshop (and repeated with similar results using ImageJ), images' open pores were selected with a color selection tool and the pixel proportions in histogram determined vs total pixels. **Figure 3.7** shows determined percent openness of the samples. MM40 had a much higher percent open than either coated or uncoated A25. It is also important to note that adding the Starbrite hydrophobic coating may reduce percent openness. Typically, a higher percent openness resulted in faster kinetics: sorbent could adsorb/desorb CO_2 and dry quicker. A shorter drying/adsorb period means a shorter full cycle time necessary in the system and more cycles possible per day. More cycles will lead to less sorbent, packets, and length of belt necessary in the system. Recent soak tests

indicate a ~90 minute drying time with a 30 minute desorb/wet time. As we find more optimal mesh samples, we will modify the system design to accommodate shorter cycle times.